

Sundeep Kumar Singh

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PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience building **scalable distributed systems** and **AI-powered platforms**. Skilled in designing high-throughput APIs with Java and Spring Boot, building production AI agents, multi-agent SDLC workflows, and RAG-based GenAI platforms. Strong experience with Kafka, PostgreSQL, Kubernetes, and Azure cloud-native microservices, with a focus on system design, scalability, and shipping end-to-end features at scale.

TECHNICAL SKILLS

Languages: Java, Python, SQL
Backend & Frameworks: Spring Boot, Spring Data JPA, Hibernate, Temporal, Django REST Framework, LangChain
Databases & Messaging: PostgreSQL, Redis, MongoDB, Apache Kafka, MinIO
AI/GenAI: RAG Architectures, LLM API Integration, Azure AI Search, Semantic Retrieval, MCP
Cloud & DevOps: Microsoft Azure, Kubernetes, Docker, GitLab CI/CD, Datadog
Other: REST APIs, Microservices, Distributed Systems, Event-Driven Architecture, Multithreading, Concurrency, Maven, Git, JUnit 5, Mockito

EXPERIENCE

Consultant (Software Engineer I) | Mercedes-Benz Research and Development India Aug 2023 – Present
Mercedes me Charge — EV Charging & Registration Platform Bangalore, India

- Architected and delivered 3+ **Spring Boot** microservices processing **5M+ events/day** for EV onboarding and charging across EMEA, AMAP, and China with full ownership of API design and schema modeling. Engineered global search using **JPA Specifications** and **materialized views**, cutting latency from 1.2s to ~120 ms (~**90%**). Overhauled Python batch jobs via cursor-based pagination, streaming, and selective parsing, reducing memory by ~**75%**.
- Designed **Temporal**-based workflow orchestration with mTLS connectivity, standardized retry logic, and a workflow staging framework. Built a reusable encrypted logging library for HTTP and Feign clients, and an **AI Agent** using **OpenAI Agents SDK** for LLM-based log summarization, reducing manual analysis by ~**70%**.
- Engineered a scalable **Kubernetes**-native job execution framework for Python workloads with secure CronJobs, adhoc execution, and parameterized Helm templates, eliminating ~**70%** of manual operational overhead.
- Spearheaded a multi-agent AI workflow with custom agent definitions and skill configurations across role-based SDLC paths with human-in-the-loop decision making, leveraging **MCP** integrations with Jira, Confluence, GitHub, and GitLab for plug-and-play AI-assisted development, increasing team delivery rate by ~**40%**.

NeoAssist — AI & GenAI Engineering Platform

- Architected an internal **LLM + RAG-powered** SDLC platform for code generation, test case generation (Cucumber, Playwright, TestNG), and code wiki — using Django, LangChain, **Azure AI Search**, and Celery, reducing manual testing effort by ~**70%**.
- Engineered **PingID MFA** authentication and built logging middleware for **Datadog** usage metrics, enabling real-time adoption monitoring and onboarded **100+ teams**.

PROJECTS

CodeForgeUI | AI-Powered Frontend Application Generator | GitHub 2026

- Architected a distributed backend using **Java 21**, **Spring Boot 4**, and **Spring AI**, enabling real-time AI code generation via SSE streaming with OpenAI, backed by Kafka Saga orchestration for exactly-once idempotent file persistence.
- Built a high-throughput API Gateway using **Spring Cloud Gateway** with Redis token bucket rate limiting (1,200+ req/s), stateless JWT authentication, Eureka service discovery, and centralized Spring Cloud Config.
- Designed a **Kubernetes live preview system** with pre-warmed pod pools, Redis distributed locks, automated MinIO file sync, and dynamic subdomain routing — delivering isolated deployments in **<30 seconds**.
- Integrated **Stripe subscription billing** with full webhook lifecycle, plan-based daily token enforcement flushed to PostgreSQL, and polyglot persistence across PostgreSQL, MongoDB, MinIO, and Redis.
- Engineered a full **observability stack** with Micrometer metrics (AI latency p50/p95/p99, Kafka saga rates), distributed tracing via **OpenTelemetry + Zipkin**, and Grafana dashboards with Prometheus.

EDUCATION

JSS Academy of Technical Education 2019 – 2023
Bachelor of Engineering in Computer Science CGPA: 9.06